

32ch portable system as flexible platform for research in biomedical ultrasound

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Background, Motivation and Objective

Research in biomedical ultrasound requires access to the transmit and receive data pipeline of electronics in order to be able to investigate new approaches. In particular, unprocessed data (e.g. pre-beamformed channel data) must be made available, so that new beamforming algorithms can be optimized and investigated. A number of multichannel research platforms have been developed over the last decade by academic or industrial groups, which fulfil these requirements. However, for some applications e.g. those related to long-term data collection or for data acquisition during proband motion, portability is an additional requirement. For this reason, we developed MOUSE, a 32ch ultrasound research system with full access to the transmit signal and receive data pipeline bridging the gap between highly compact wearable systems (e.g. WULPUS, TinyProbe) and flexible high-performance research scanners (e.g. Verasonics, ULAOP).

Statement of Contribution/Methods

The system allows parametrization of signals using tri-state transmit ICs combined with PWM in a frequency range of 100 kHz to 10 MHz. Receive signals are amplified with up to 44.3 dB at a sampling rate of up to 50 MSPS and 12 bits. For data management, communication and sequence control, a ZYNQ 7 FPGA is available. The system is typically driven from a windows operating system using different UIs or an SDK, however, an iOS interface is available as well. Data transfer is realized using USB 3.0 in the cabled setup or in Wi-Fi mode with a data transmission rate of approximately 15 Mbit/s. The MOUSE can be powered using a 12V medical power supply or with a battery in the portable configuration.

Results/Discussion

The MOUSE system has been characterized and used in different medical applications. The portable system implementation (with battery and Wi-Fi mode) has been used together with a customized linear array for tracking of muscle motion during exercise as a biofeedback for physiotherapy training. The 256 element probe with integrated 8:32 multiplexer ICs allows recording of thickness variation of the TrA (Transversus abdominis) in proband experiments. In another implementation, the MOUSE system was used for automatic estimation of the bladder volume when driving a customized (longterm attachable) 32-element linear array. The probe was driven in plane wave compounding mode and compounded B-scan images (13 angles each) were acquired and reconstructed with 23 frames/s. Finally, in a third application, the MOUSE was used for tracking of muscle motion. Customized transducer bracelets (32 elements, 1 MHz and 4 MHz) were connected to the system and features being characteristic for defined hand postures were extracted from the A-scan signals. It could be shown that hand motion in up to 3 degrees of freedom could be correctly recognized in the ultrasound data.

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